

A course in language creation



Conlangs University

Information Structure



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Miacomet, Tutor

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A lot of conlanging focuses on standalone sentences, if not on sounds and individual words. We end up missing a lot of the subtleties of language that occur in context during conversation. This article will focus on some of the ways that information can be presented differently in various contexts. It's painted in broad strokes, which I hope will be helpful to conlangers, but I'll give some citations at the end to look at if you want to dive deeper. This article draws particularly on [Aissen \(to appear\)](#).

What is information structure?

Information structure is how the information in a particular sentence relates to the wider context. Most of the time, when we're using language, it's to communicate with specific people in specific contexts. When we speak, we make assumptions about what sorts of information our listeners have, and we organize our speech according to that.

The **common ground** (CG) refers to the set of all information that the speaker and listener share. This can contain general knowledge about the world, knowledge the speakers know they share (e.g. domain knowledge or knowledge from past experiences), and the information that's already been added during the conversation. Most of the time, statements add information to the common ground and questions request that whoever you're talking to add a specific piece of information to the common ground.

No conversation starts from scratch. Most of the time, things you say rely on other things being held in the common ground. For example if you ask someone "when did you first meet Paul?" that presupposes that they *have* met Paul. The speaker has to think that the knowledge that the listener has met Paul before is in the common ground (but that the time they met him is not) for that to make sense as a question in context.

In addition to all of the facts, or propositional content, in the common ground, there's a set of known people, places and things, whose existence has been established. If I say something about "the orange cat," that presupposes that somewhere in the common ground, there's an orange cat that both of us can pick out as the thing that my statement refers to.

If the goal of a statement is to add information to the common ground, then it makes sense that most statements will contain some new information. At any point in a discussion, you can imagine there being a **question under discussion** (QUD) which someone is trying to answer. Sometimes this is a literal, overt question. If you ask me "when did you first meet Paul?" then I might answer "I first met Paul two winters ago." The QUD was "when did you first meet Paul" and I answered it by providing the information that I met him "*two winters ago*." Sometimes it's not an overt question. If I follow the last sentence with "I met Paul through mutual friends," then I'm answering the question "how did you meet Paul?" You can think of conversations as moving through different sets of questions, implicit and explicit, as different pieces of information get added to the common ground.

It's important to note that the same information can answer different questions in different contexts. "I texted Paul to come back" could be an answer to "Who did you text to come back?" or "Who texted Paul to

come back?" or "What did you text Paul?" or "How did you tell Paul to come back?" Even vague questions like "What did you do?" or "What happened?" Often, that same information will be packaged into a sentence differently, depending on what question the speaker's trying to answer.

What can get marked?

The specific piece of information that answers the question under discussion is the **focus**. Often, the focus will be some new piece of information that the speaker is providing (in contrast with the background, the rest of the sentence). If someone asks "what did Roy see?" and you answer "Roy saw the painting," then the piece of information answering the question, the focus, is "the painting." Your conversational partner knows Roy saw *something*, so that's in the background, and you're adding that what they saw is the *painting*.

Depending on what sort of context a sentence is uttered in, different parts of it can be focused. Each of the examples in (1) show a possible question that the sentence "Roy saw the painting" could answer. The focus in each context is enclosed in [brackets]. You can focus a single word or the entire sentence. Sentences like (1.c), where the verb and everything after it is focused are said to have **predicate focus** and ones like (1.d), where the whole sentence is focused, are said to have **broad focus**.

1. a. What did Roy see?
Roy saw [the painting]
- b. Who saw the painting?
[Roy] saw the painting
- c. What did Roy do?
Roy [saw the painting]
- d. What happened?
[Roy saw the painting]

These differences don't show up in isolated utterances, but only make sense in context! That's why when you're conlanging, it's important to think about dialogues and stories, not just single example sentences.

Another way to think about focus is that focusing a particular element of a sentence evokes a set of alternative sentences that can be made by replacing the focused phrase with other alternatives. Sentence (1.b), with focus on "Roy," specifically contrasts with other sentences where "Roy" is replaced with other alternatives, like Anna or May. The sentence "Roy saw the painting" would be said in a context where it contrasts with "Anna saw the painting" but probably not in one where it contrasts with "Roy saw the *sculpture*."

There are some expressions, like *only* or *also*, that interact with focus. In English, *only* makes a focus expression **exhaustive**: it says that not only is the statement true, but all of its focus alternatives are false. Saying "Chi only ate *cake*" (with focus on "cake") tells you that "Chi ate cake," is true but also that replacing "cake" with any other reasonable alternative is false (Chi didn't eat pie or cookies). "Cake" is an exhaustive list of what Chi ate.

On the other hand, *also* makes a focus expression **additive**. Saying "Chi also ate *cake*" tells you that in addition to "Chi ate cake" being true, there's some other sentence where replacing "cake" with a reasonable alternative is true. This sort of sentence makes sense in a context where an alternative like "Chi ate cookies" is already part of the common ground, and you're adding that "Chi ate cake" is true in addition to its alternative.

Another important category in information structure is the **topic**, something in the common ground that sets what the sentence is about. When you're communicating, it can be helpful to link information you're adding to the common ground with information that's already in the common ground. The topic serves as the anchor for the **comment**, what you're saying *about* the topic, to be added to the common ground. Since the topic is part of the common ground, it has to be a given. It can't be new information, so the topic and the focus are never the same. It's worth noting that just because something isn't a focus doesn't mean it's a topic, and just because something isn't a topic doesn't mean it's a focus. It's completely normal for sentences to have a topic, a focus, and other stuff that's neither topic nor focus.

A topicalized phrase used specifically to mark a difference in topic between two clauses is a **contrastive topic**. Contrastive topics bring something given, but backgrounded, to the foreground, often to show some difference between the previous topic and the new one. An example of a contrastive topic is Max in "Pedro texts me every day. Max, about once a week." In the second sentence, the speaker changes the topic from Pedro to Max, with the comment "[texts me] about once a week."

Some languages distinguish another type of topic, which is given and central to the information, but not contrastive and completely recoverable from context (and often omitted). When these *do* show up, they're sometimes called **antitopic** and often occur at the ends of clauses, as a sort of afterthought.

Some languages also have a category sometimes called **frames**, which give context for the interpretation of a sentence and tend to get treated similarly to topics, but aren't really what the sentence is *about* (you might call that an 'aboutness topic'). An example is in the sentence "In New York, the cost of rent is very high," where the phrase "in New York" gives context for the utterance and acts in some ways like a topic, but the sentence is really *about* "the cost of rent." Some languages allow sentences to have both a frame and an aboutness topic marked as topics, but wouldn't allow two aboutness topics, so it can be useful to distinguish them.

How does it get marked?

Okay cool, that's a lot of theory and a lot of different words. But why does a conlanger care about this? Natural languages have a variety of different ways to mark these categories in their grammars. They're not often taught explicitly in school, and they can be easy to miss in your native language. It's also easy to accidentally bring them with you from your native language to others.

This section is going to give an overview of a few common ways that information structure can be marked and give examples from different natural languages. I'll also talk about how these strategies interact with each other and with a few other aspects of language. This is by no means exhaustive, but I hope it'll give inspiration for a few different things to think about while conlanging.

The most straightforward way you might see information structure marked is morphologically. Morphological IS marking could be as simple as a morpheme that marks topic or a morpheme that marks focus. Here are some examples of **topic markers** and **focus markers** from Miyako (Ryukyuan, Japonic).

2. unu katana=se =e skss-ai -n
PROX knife =INS=TOP cut -POT-NEG
"I can't cut it with this knife."
3. pav =nu =tu faa =iu ff -i
snake=NOM=FOC child=ACC bite-CVB
"A snake bit the child."

Sometimes these morphemes are sensitive to other grammatical properties. For example, Miyako has a topic marker specifically for direct objects.

4. ansi=tu tauf=fu =pa kff
so =FOC tofu=ACC=TOP.DO make
"This is how we make tofu."

Just like other categories, you can have suppletive forms that are used when topicalized or focused.

5. araa tin =nu =tu nustu =n nisum-as -i
1SG.TOP money=ACC=FOC burglar=DAT steal-CAUS-CVB
"I got my money stolen by a burglar."

In addition to marking on constituent that *is* focused, sometimes there's marking on constituents around the one that's focused. One form this can take is **focus agreement**, where a verb agrees with its arguments differently depending on whether they're focused or not. Here's an example from Yukaghir (Isolate). When the direct object of the verb is focused, it takes a specific set of agreement suffixes.

6. Tas köde metin cogojə-ləs tadi-mələ
that man to.me knife -FOC give-OBJ.FOC.3SG
"That man gave me a knife."

Another phenomenon, sometimes called **focus concord** or by its Japanese name ***kakari-musubi***, occurs when the presence of a focused element forces verbs to take a specific form, different than the form they'd usually take. Here are some examples from Sinhala (Indo-Aryan, Indo-European).

7. a. mamə ē potə kiyewwa
 1SG that book read
 'I read that book.'
- b. mamə ē potə-y kiyeww-e
 1SG that book-FOC read -FOC
 'It's that book that I read.'

The default past-tense verb in Sinhala ends in **-a**. But when something in the clause is focused, the focused constituent takes a **-y** suffix and the verb takes the ending **-e**. When systems like this do exist, it's relatively common for the forms triggered by focus to look like forms used in subordinate clauses. One theory is that these come from cleft constructions, like "is *this* what you'll take?" which a lot of languages use to mark focus periphrastically.

Prosody

Many languages give some sort of prosodic prominence to the focus. You might do this by putting a certain intonation or stress on the focus or by requiring the focus to be in a certain prosodic position.

English marks the focus by putting sentence-level stress on the last stressed syllable of the focus. Thinking back to example (1), you can get focus marking on different parts of the same sentence by asking different questions. Interestingly, the order of the words never changes, just the intonation.

Take a look at these examples, where I marked sentential stress with an acute accent. To focus the object, you put sentential stress on the last stressed syllable of the object, to get "Roy saw the páinting." To focus the subject, put it on the subject, and so on.

8. a. What did Roy see?
 Roy saw [the páinting]
- b. Who saw the painting?
 [Róy] saw the painting
- c. What did Roy do to the painting?
 Roy [sáw] the painting
- d. What did Roy do?
 Roy [saw the páinting]

Sometimes, like in (8.d), you get a whole phrase in focus. You still put sentential stress on the last stressed syllable of the phrase, but sometimes that'll create an ambiguity: even though the scope of the focus in (8.a) and (8.d) is different, the intonation ends up being the same! That's fine, it never stops English speakers from understanding each other!

In Chichewa (Bantu, Niger-Congo), the focus has to be on the right edge of a prosodic phrase. Chichewa speakers lengthen the second-to-last vowel in a prosodic phrase, so you can split them apart pretty easily. In (9.a) you can see that the whole sentence is a single prosodic phrase. The only long vowel is the u in

Mavuto, at the end of the sentence. To answer the question in (9.a), you probably want to focus "mnyumbá yá Mávúto." It's already at the right edge of a prosodic phrase, so you're good to go!

On the other hand, if you're answering the question in (9.b), you want to focus "anagóna" but *not* "mnyumbá yá Mávúto." That's a problem, because "anagóna" doesn't start out at the right edge of a prosodic phrase. Chichewa grammar lets you fix it easily though: you can pop in a prosodic break after the phrase you're focusing. This shows up as a slight pause and a lengthening of the second-to-last vowel. You can see that characteristic long vowel in "anagóona" in (9.b).

9. a. Q: Where did they sleep?
A: anagóna m- nyumbá yá Mávúuto
they.slept in-house of Mavuto
"They slept in Mavuto's house"
- b. Q: What did they do in Mavuto's house?
A: anagóona m- nyumbá yá Mávúuto
they.slept in-house of Mavuto
"They slept in Mavuto's house"

Both of these are purely prosodic systems. The prosody changes, but the words themselves aren't affected. As you'll see in the next section, sometimes prosody interacts with other things to make more complicated systems. Another thing you can see is that both of these systems have some ambiguity, but that's okay! When you're making a naturalistic conlang at least, it's better to create a system that works well with the existing prosodic structures of your language than to try and contrive something that'll be completely unambiguous.

Word Order

Word order very commonly reflects information structure. When someone says a natural language has "free word order" or that its words are ordered "by importance," that pretty much always really means that information structure plays an important role in word order. But even with topic-prominent languages, there is a default ordering of constituents that you might see if you topicalize something like a time phrase, leaving the S, O, and V in place. Classical Latin is said to have free word order, but it still defaults to SOV order in cases of broad focus or predicate focus. If you're trying to make a "free word order" conlang, then try and think about it in terms of topics and focuses instead!

One common setup is to have topics or focuses come at the beginning of a clause. Languages that consistently start sentences with topics (regardless of their role in the sentence) and follow them with the rest of the information are called **topic-prominent** languages and are said to have **topic-comment** structure.

The most well-studied topic-prominent languages tend to be those from East Asia, so here are a few examples from Cantonese (Sinitic, Sino-Tibetan). Topics can be constituents in the sentence that got moved up to the topic position, like in (10), where the object of the verb "know" gets topicalized. They can have a relationship with a later part of the sentence to give context, like how "beer" in sentence (11) sets a context for the listener to understand "what kind" in. They can also set a general frame to understand the sentence in, like in sentence (12).

10. lī dī yéh móuh yàhn sīk ge
 this CL thing not.have person know DP
 "This sort of thing, nobody knows."
11. bējáu léih jeui jūngyī bīn jek a
 beer you most like which CL Q
 "As for beer, what kind do you like the most?"
12. Hēunggóng làuhga gwai dou séi
 Hong Kong flat.price expensive until die
 "In Hong Kong, flat prices are ridiculously expensive."

Even outside of topic-comment languages, it's common to have some kind of IS position at the beginnings of sentences. If you're familiar with any V2 languages like German, Dutch or any Scandinavian lect, you might recognize that whatever goes before the first verb tends to be either a topic or a focus. Another possibility is contrastive topic extraction. Some languages, like English, let you pull topics out to the front of the sentence, but only if they're contrastive. Without context, "the cake I liked" feels odd. OSV word order? In *my* English? It's more likely than you think: suppose you had some desserts and liked the cake, but not the pie. Your friend says you liked the pie, but they're wrong. You correct them: "The *cake* I liked. The *pie* I didn't like." English allows this, but only for specific kinds of topic, not for all.

Many languages also front focus phrases. Generally if a language allows wh-movement, then fronted focus phrases will move to the same place as wh-words. Similarly if a language doesn't allow you to move wh-words out of certain constructions or requires the wh-words to bring other words along with them when they move, that will tend to be true of moved focus phrases as well. Here's an example of focus fronting from Tzotzil (Mayan) where the object of the verb "they found" gets moved to the beginning of the sentence.

13. tseb San-Antrex la te staik un
 girl san andrés CL there they.found PAR
 "It was a San Andrés girl they found there."

Kind of like topic-prominent languages, you can also have focus-prominent languages. Native North American languages in the Pacific Northwest sprachbund tend to structure sentences so that the first thing in the sentence is the focus, especially informational focus, and is followed by the topic. (You'll sometimes see the informational focus called the **rheme** and the rest of the phrase called the **theme** in literature on these languages.)

In St'át'imcets (Salishan, also called Lillooet), the first thing in a clause is the predicate. Most lexical words can be predicates and most lexical words can be arguments of those predicates, so languages like St'át'imcets are said to have very weak distinctions between classes like noun, adjective and verb. Whenever possible, the predicate of the sentence should be new information. In (14.a) the topic is "the coyote" and the focus is the predicate, "goes along." In (14.b) on the other hand, the coyote is focused and moves to the predicate slot, ending up with something like "the-going-along-one is a coyote."

14. a. t'ak ti= nk'yáp=a
 go.along DET=coyote=DET
 "The coyote goes along."
- b. nk'yap ti= t'ák =a
 coyote DET=go.along=DET
 "The one who goes along is a coyote."

Sometimes prosody interacts with word order. We've already seen examples where prosodic features move around to match the position of the focus. What about where the focus moves around to match prosodic features?

In Basque (Isolate), the focus is in a prosodically prominent position immediately before the verb. If the focus is already right before the verb, then it can receive focus stress like usual, like in (15.b). If you want to focus some other part of the sentence, you can't do it without changing the word order. You have to move whatever's focused to come right before the verb, where it can take the focus stress in that position. In (15.c), the subject "Mirenek" moves after the object to take on that pre-verbal position and allow for focus stress.

15. a. Q: What happened?
 A: Miren-ek Jon ikusi rau
 Miren-ERG Jon see AUX
 "Miren saw Jon"
- b. Q: Who did Miren see?
 A: Miren-ek Jon ikusi rau
 Miren-ERG Jon see AUX
 "Miren saw Jon"
- c. Q: Who saw Jon?
 A: Jon Miren-ek ikusi rau
 Jon Miren-ERG see AUX
 "Miren saw Jon"

This pre-verbal focus position turns out to be pretty common among verb final languages, and is attested in SOV languages from diverse families including Indo-European, Turkic, and Sino-Tibetan. On the flip side, some Bantu languages have a prosodically specified focus position immediately *after* the verb.

Many languages also have periphrastic constructions that can be used to structure information. A common example of this is **cleft sentences** which use the verb "to be" to link a subject with a subordinate clause that describes the subject. (16) gives two examples of clefts, both of which are clefted versions of the sentence "Paul saw me on Monday."

16. a. It was Paul who saw me on Monday.
 b. It was on Monday that Paul saw me.

In (16.a), you're presupposing that the relative clause "who saw me on Monday" identifies someone in the common ground. You and the listener both know someone saw you on Monday, but don't know who, so you identify them as Paul. In (16.b) on the other hand, you're presupposing that Paul saw you, and adding the information that it was on Monday. Ringing any bells? Clefts act very similar to focus expressions, and may be a more common way of expressing the same semantics. (If you look back, you'll see some examples of focus, like (7.b) (13) and (14.b) were translated to English using clefts.)

Interactions with other Features

Nothing in a language acts alone. Information structure is no exception. When you add a new feature to a language or want to flesh out an existing one, you should think about how it can interact with other bits of your language. Here are a few other things to think about

Question words often behave as though they are focused. Question words may be marked as focus, and languages that allow wh-movement often also allow focus movement to the same position. Example 17 shows a question word taking focus marking in Miyako and example 19 shows a question word in Palauan (Oceanic, Austronesian) taking the same "patient focus" agreement triggered by fronting the focused patient in 18.

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17. a =ka asta=a nta =n =tu aw
    1SG=NOM clog=TOP where=DAT=FOC be
    'Where are my clogs?'

18. a ldioui a- l- lirell-ii a Naomi el mo er ngak
    dessert IR3-PF-make -3sg Naomi LK go P me
    "Naomi made a dessert for me."

20. ng-ngera a- l- silseb-ii a se'el-il
    CL-what IR3-PF-burn -3sg friend -3sg
    "What did his friend burn?"
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Information structure also interacts with **definiteness**, since both serve similar functions of tracking what is or isn't in the common ground. Nouns can get marked as definite if they pick out something specific and unique in the common ground. You can only say "the cat" if there's already some cat in the common ground that you expect your listener will be able to identify as *the* cat. If something is readily identifiable even without context, you can still mark it as definite without introducing it in conversation, because it's already in the common ground. Saying "the sun" or "the moon" leaves little doubt what you're talking about, at least on my home planet, the Earth, where there's only one of each. Since topics have to be part of the common ground, noun phrase topics tend to be definite.

On the other hand, **indefinite** nouns are *not* part of the common ground. Since they aren't part of the common ground, they can't occur as topics, and are more likely to occur as focuses. However, it's not one-to-one: it's common for focuses to be definite, such as back in example (1) where nouns like "Roy" and "the painting" definitely pick out unique, familiar referents, but can still get focused when they're used to provide new information.

Rather than being topic-prominent, many languages are **subject prominent**, meaning that they give special prominence to a particular verb argument, usually the S of intransitive verbs and the A of transitive verbs. In some subject-prominent languages, there's a tendency to make the topic the subject whenever possible. This can be done by choosing how to frame the sentence or through operations like the passive voice, which can promote a patient to subject. English does this. If you ask someone, "What happened to Crystal?" then they could reply something like "Crystal left the house and got eaten by a lion." In a sentence like this, "Crystal" is the topic and what happened to her is the focus, like you'd expect. The second clause is equivalent to "A lion ate Crystal," but to keep the topic and subject the same between the two clauses, you can use the passive voice.

Conclusion

This intro might seem long, but I promise you: it barely scrapes the surface. There's a lot to learn about information structure, and a lot of really interesting open questions that linguists are still learning the answers to. I hope this will serve as a jumping-off point to learn about the different sorts of possibilities and will get you thinking about how your conlang structures information. Writing this definitely got me thinking.

I do think there are a few takeaways other than just introducing things like topic and focus: Beware "free word order," most of the time I see it in conlangs, it's code for either "pragmatically motivated word order" or "I'm not thinking very much about word order." Don't create all your example sentences in vacuums! When you're thinking about this kind of thing, it's useful to situate sentences within contexts, as parts of dialogues, or inside larger narratives. In your conlang documentation, you can give a sentence about the context to make it clearer! And just because something doesn't show up in writing or segmental phonology doesn't mean it's not important. The English and Chichewa examples show the power of prosody!

Happy conlanging! I'm excited to see what information you add to the common ground.

Citations

If you're curious about any of the examples or want to learn more than my intro, I've included a few sources that I used while putting this together, including sources for all example sentences other than the English ones. I recommend starting out with Aissen (To appear) and branching out from there.

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